

Question ID: 3b225698

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

Triangle XYZ is similar to triangle RST such that X , Y , and Z correspond to R , S , and T , respectively. The measure of $\angle Z$ is 20° and $2XY = RS$. What is the measure of $\angle T$?

Answer

- A. 2°
- B. 10°
- C. 20°
- D. 40°

Correct Answer: C

Rationale

Choice C is correct. It’s given that triangle XYZ is similar to triangle RST , such that X , Y , and Z correspond to R , S , and T , respectively. Since corresponding angles of similar triangles are congruent, it follows that the measure of $\angle Z$ is congruent to the measure of $\angle T$. It’s given that the measure of $\angle Z$ is 20° . Therefore, the measure of $\angle T$ is 20° .

Choice A is incorrect and may result from a conceptual error.

Choice B is incorrect. This is half the measure of $\angle Z$.

Choice D is incorrect. This is twice the measure of $\angle Z$.